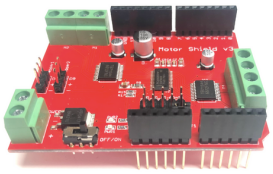
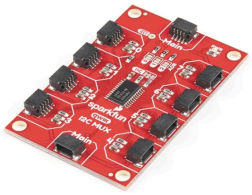
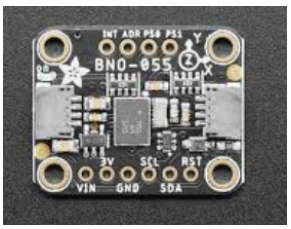
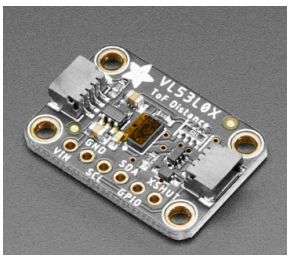
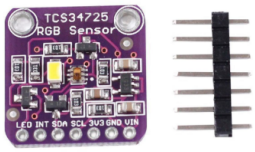
| | | | |
|---|---|---|---|
| Carobot V3  | Motor Shield | Issues:<br>Is connected to I2C( Need to jump cables)<br>Has an allcall address of 0x70, which conflicts with the multiplexer. | Documentation<br>https://learn.carobot.ca/guide/motor-shield-v3 |
| TCA9548A qwiic mux  | I2C multiplexer | Address conflicts with Carobot.<br>Pins get easily deformed | https://cdn.sparkfun.com/assets/f/0/4/b/3/tca9548a.pdf |
| BNO055 absolute orientation sensor  | IMU | The sensor has no reset: the heading is always global.<br>The sensor has a 360 degree wraparound, which is sensitive to fluctuations. | https://cdn-shop.adafruit.com/datasheets/BST_BNO055_DS000_12.pdf<br>code:https://adafruit.github.io/Adafruit_BNO055/html/class_adafruit___b_n_o055.html |
| VL53L0X TOF sensor  | TOF Lidar | The sensors I2C address is 0x29, which conflicts with TCS34725.<br>Can't handle below 30mm of range | https://learn.adafruit.com/adafruit-vl53l0x-micro-lidar-distance-sensor-breakout/overview |
| TCS34725  | Color sensor | Its unchangeable address conflicts with Vl53l0X.<br>Measurement takes 50ms | https://adafruit.github.io/Adafruit_TCS34725/html/class_adafruit___t_c_s34725.html |

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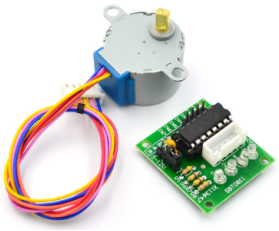
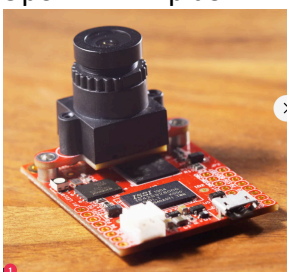
|---|---|---|---|
| <br>195:1 Metal Gearmotor 20Dx44L mm 12V CB with Extended Motor Shaft | Motor | **Extended shaft for encoder** | https://www.pololu.com/product/3493 |
|  | Stepper motor | The motor itself is 5V, so you need a 5v buck. | https://lastminuteengineers.com/28byj48-stepper-motor-arduino-tutorial/ |
| OpenMV H7 plus.<br>  | Camera | Memory can leak from power brownouts without and SD card. | https://docs.openmv.io/library/omv.image.html |
| | Lipo battery | ALWAYS unplug the robot before charging. | |